AS-Level 9709 Paper 1/2/4/5 Diagnostic Readiness

PRACTICE WORKSHEET

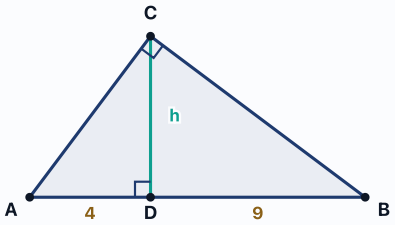
Pure 1, Mechanics & Statistics Readiness • Cambridge 9709

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AS-Level 9709 Paper 1/2/4/5 Diagnostic Readiness Practice Exercises

1 MCQ

The diagram shows a right-angled triangle ABC, with the right angle at D. The altitude from D to the hypotenuse BC is drawn. Given that BD = 4 cm and DC = 9 cm, find the length of the altitude AD.



A 6 cm

B 5 cm

C 7 cm

D 8 cm

2 MCQ

A fair six-sided die is rolled 4 times. Let X be the number of times a 6 is obtained. The possible values for X are 0, 1, 2, 3, and 4. The probability of obtaining a 6 on any given roll is $p = \frac{1}{6}$. The number of trials is $n = 4$. This scenario can be modelled by a binomial distribution, denoted as $X \sim B(n, p)$. Which of the following represents the correct parameters for this binomial distribution?

A $B(4, 1/6)$

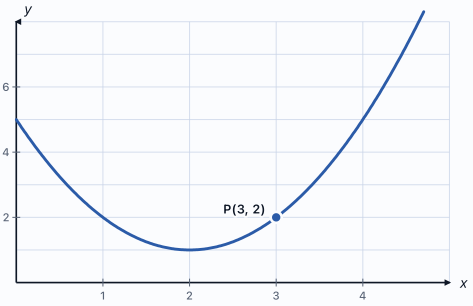
B $B(6, 1/4)$

C $B(1, 4/6)$

D $B(4, 4/6)$

3 MCQ

The diagram shows part of the curve with equation $y = x^2 - 4x + 5$ and the point $P(3, 2)$ lying on the curve. Find the gradient of the tangent to the curve at the point P.



A 2

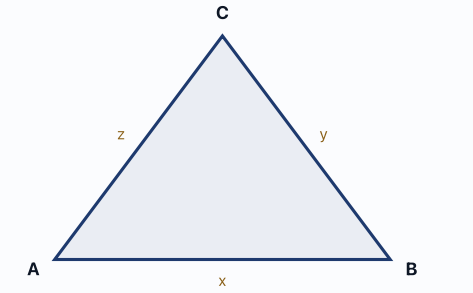
B 6

C -4

D $\frac{2}{3}$

4 MCQ

The diagram shows a triangle ABC with coordinates A(2, 1), B(8, 1) and C(5, 5). Find the length of the side AC.



A 5

B $sqrt26$

C 6

D 7

5 MCQ

In a class of 20 students, 8 study only French, 7 study only Spanish, 3 study both, and 2 study neither. A student is picked at random. Find the probability that the student studies French.

A 1/2

B 4/5

C 7/8

D 11/20

6 MCQ

Examining a class of 29 students, 5 study only French, 3 study only Spanish, 3 study both, and 18 study neither. A student is picked at random. Find the probability that the student studies French.

A 1/2

B 8/29

C 3/4

D 3/7

7 MCQ

Examining a class of 24 students, 6 study only French, 8 study only Spanish, 5 study both, and 5 study neither. A student is picked at random. Find the probability that the student studies French.

A 2/7

B 11/24

C 1/3

D 1/2

8 MCQ

Find the mean of the data set: $22, 5, 24, -5, 25, 22, -9, 28$.

A 20

B 15

C 14

D 9

9 MCQ

Events A and B are such that $P(A) = 3/5$, $P(B) = 1/5$, and $P(A \cap B) = 1/10$. Find $P(A \cup B)$.

A 7/10

B 11/15

C 4/5

D 4/7

10 MCQ

Calculate the median of the data set: $1, 5, 10, 35, 39, 40, 42, 49$.

A 37

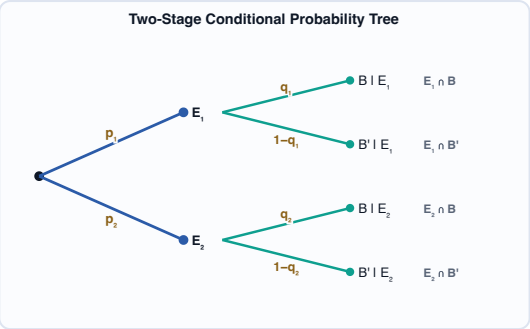
B 32

C 39

D 40

11 MCQ

In an applied problem, a bag contains 15 balls, of which 8 are red. Two balls are drawn at random, one after another, without replacement. Find the probability that both are red.



A 9/14

B 8/11

C 4/15

D 7/9

12 MCQ

Events A and B are such that $P(A) = 2/5$, $P(B) = 2/5$, and $P(A \cap B) = 1/5$. Find $P(A \cup B)$.

A 3/5

B 1/2

C 6/7

D 7/11

13 MCQ

A particle moves along a straight line. Its velocity, v m/s, at time t seconds is given by $v = 3t - 2$. Find the displacement of the particle during the first 4 seconds of its motion.

Given Motion Parameters

Parameter 1	Parameter 2
2	4

A 20 m

B 16 m

C 24 m

D 28 m

14 MCQ

The two-way table shows the results of a survey of 100 students on their preferred mode of transport to school and their year group.
What is the probability that a randomly selected student is a Year 1 student who prefers to travel by car?

	Car	Bus	Walk
Year 1	15	20	10
Year 2	25	18	12

A 0.45

B 0.33

C 0.38

D 0.15

15 MCQ

A particle moves in a straight line such that its displacement s metres from a fixed origin after t seconds is given by the equation $s = t^3 - 6t^2 + 9t + 2$.

The velocity of the particle is given by $v = \frac{ds}{dt}$.

Find the value of t for which the particle is instantaneously at rest and then determine the displacement of the particle at that instant.

A t=1 s, s=6 m

B t=3 s, s=2 m

C t=2 s, s=4 m

D t=1 s, s=2 m

16 MCQ

A particle's velocity v m/s at time t seconds is given by $v = \frac{1}{2}t^2 - 3t + 5$. The displacement of the particle from a fixed origin is s metres. It is known that at $t = 2$, the displacement $s = 7$. The displacement function $s(t)$ can be expressed in the form $s(t) = At^3 + Bt^2 + Ct + D$. Find the value of A and the value of D.

Given Motion Parameters

Parameter 1	Parameter 2
1	2

A A = 1/6, D = 10/3

B A = 1/3, D = 5/3

C A = 1/6, D = 5/3

D A = 1/3, D = 10/3

17 MCQ

A research team is studying the impact of different fertilizer types on crop yield. They conducted an experiment with two fertilizer types, A and B, and recorded the yield in kilograms per plot. The data is summarized in the two-way table below. The total number of plots is 200.
Given that a plot was selected at random and its yield was less than 50 kg, what is the probability that it received fertilizer type B? Provide your answer as a fraction in its simplest form.

Fertilizer Type vs. Crop Yield

Fertilizer Type	Yield < 50 kg	Yield ≥ 50 kg	Total
A	30	70	100
B	50	50	100
Total	80	120	200

A 1/2

B 3/8

C 1/4

D 5/8

18 MCQ

Solve $x^2 - 2x - 8 = 0$.

A x=-2 or x=5

B x=-3 or x=4

C x=-2 or x=4

D x=2 or x=-4

19 MCQ

Rewrite in fully simplified form $p^3 \div p^2$.

A p^5

B p^0

C p^1

D p^2

20 MCQ

Express in simplest algebraic form $\sqrt{396}$, giving your answer in the form $a\sqrt{b}$ with b as small as possible.

A $6\sqrt{44}$

B 66

C $6\sqrt{11}$

D $6\sqrt{12}$

21 MCQ

Rationalise the denominator of $\frac{6}{\sqrt{19}}$.

A $\frac{\sqrt{19}}{19}$

B $\frac{6\sqrt{19}}{19}$

C $\frac{6\sqrt{19}}{6}$

D $6\sqrt{19}$

22 MCQ

Write $a^{-3} \times a^{-1}$ in the form a^k, giving the value of k.

A k=-3

B k=4

C k=-4

D k=-2

23 MCQ

Express $x^2 + 4x$ in the form $(x+p)^2 + q$.

A $(x+2)^2$

B $(x+2)^2 + 4$

C $(x+2)^2 - 4$

D $(x-2)^2 - 4$

24 MCQ

Rationalise the denominator of $\frac{2}{3+\sqrt{7}}$.

A $\frac{2(3-\sqrt{7})}{3}$

B $2(3-\sqrt{7})$

C $\frac{2(3+\sqrt{7})}{2}$

D $\frac{2(3-\sqrt{7})}{2}$

25 MCQ

Solve $|x-2| > 7$.

A $x < -4$ or $x > 9$

B $x < -5$ or $x > 8$

C $x < -5$ or $x > 9$

D -5<x<9